

Nicolas BRUNON

Principal AI Systems Architect | Large Scale RAG & Security

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Summary

Principal Systems Architect with 20+ years of experience specializing in the intersection of **Enterprise Security and Applied AI**. Recently engineered and deployed a production-grade **Retrieval-Augmented Generation (RAG)** system for regulated financial environments. Expert in designing **Identity-Aware AI architectures** that bridge the gap between frontier LLMs and complex enterprise data governance. Focused on the safe, secure, and scalable adoption of AI systems.

Core Expertise

AI Systems & Safety

- **Generative AI Architecture:** End-to-end RAG pipelines, LLM orchestration, prompt engineering.
- **AI Security:** Identity-aware retrieval, hallucination mitigation, LLM red-teaming/security.
- **Data Infrastructure:** Vector databases (Azure AI Search), DataLake, Azure Synapse, Spark Cluster On Demand.

Cloud & Distributed Systems

- **Platform Engineering:** Azure Full Suite.
 - **Security Design:** Zero Trust Architecture, OAuth Flows, API Governance.
 - **Enterprise Strategy:** AI adoption roadmaps, technical governance, and stakeholder alignment.
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Professional Experience

Principal AI & Cloud Solutions Architect

Candriam | 2007 — Present

Lead architect for enterprise-scale AI and cloud transformation in a highly regulated financial environment.

- **Production RAG Implementation:** Engineered a production-ready RAG assistant. Solved critical challenges regarding **retrieval precision** and **hallucination control** for operational teams.
- **Identity-Aware AI Security:** Designed and implemented a security framework ensuring that LLM responses respect granular user permissions via **Identity Propagation** across the vector search pipeline.
- **AI Strategy & Governance:** Advisor to the Executive Committee on AI adoption, balancing rapid innovation with strict compliance and risk management.
- **Infrastructure Modernization:** Orchestrated large-scale migrations from legacy on-premises systems to modern, scalable cloud-native architectures.
- **Leadership:** Proven ability to build consensus between technical teams and executive leadership on complex architectural shifts.

Distributed Systems Engineer

Vistali (Independent Consultant) | 2004 — 2007

- Provided specialized consulting in **Enterprise Service Bus (ESB)** and middleware solutions.
 - Developed complex distributed system architectures using .NET and BizTalk for high-throughput environments.
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Selected Projects

Secure Identity-Aware RAG Framework

Architected a reference implementation for secure enterprise-grade RAG systems.

- **Problem:** Preventing data leakage in LLM applications where users have different access rights.
 - **Solution:** Implemented a pipeline ensuring **identity propagation** from the user prompt to the vector database retrieval.
 - **Architecture:** User → Entra ID → Retrieval → Authorization → LLM
 - **Tech:** Python, Vector DB, OAuth2/OIDC, Azure AI.
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Education & Certifications

- **Master's Degree in Engineering** École Supérieure d'Ingénieurs de Marseille (now **Centrale Marseille**), 2003.
- **Professional Development:** Azure AI Associate, Azure Infrastructure, ITIL v3.